

Sequence-based computational algorithm can predict antimicrobial resistance in major human pathogens: A comparative analysis with traditional clinical methods

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Introduction: Antimicrobial resistance (AMR) poses a significant public health challenge, with detection becoming progressively more difficult. Conventional clinical practices rely on culture-dependent antibiotic susceptibility testing (AST) however, advancements in sequencing techniques have facilitated the emergence of sequence-based AMR prediction as a more rapid approach. In this study, we use an AMR gene classification algorithm, KARGA, to genotypically characterize AMR human pathogens and evaluate the prediction performance in comparison to traditional clinical methodologies.

Methods: Bacterial organisms resistant to multiple antibacterial agents were isolated from patients at a large, tertiary teaching hospital as part of a prospective AMR surveillance study. AST was performed using MICs by the hospital as standard-of-care. Whole-genome sequencing of the isolates was conducted using Illumina MiSeq. AMR gene prediction was employed using *k*-mer based antibiotic gene resistance analyzer (KARGA), an alignment-free tool that uses hash-based *k*-mer mapping to a reference AMR gene database (MEGARes 3.0). Percent coverage for individual genes were aggregated for each of the 27 antimicrobial drug classes in the reference database, resulting in a standardized gene coverage score for analysis. Scores for each class were summarized descriptively and compared between species using Kruskal-Wallis test. ROC analysis was conducted to evaluate the predictive performance of the gene coverage score in identifying the AST resistance pattern on a subset of isolates.

Results: Whole-genome sequences from 76 drug resistant bacterial pathogens were included in the analysis. Pathogens were *S. aureus* (n=49), *Pseudomonas sp.* (n=14), *E. coli* (n=7), *K. pneumoniae* (n=3), and *Acinetobacter sp.* (n=3). After adjusting for multiple comparisons, significant differences were observed in gene coverage scores between pathogen species for 22 out of 27 drug classes, including beta-lactams ($p<0.001$), fluoroquinolones ($p<0.001$), and macrolides ($p<0.001$). Among *S. aureus* isolates with associated AST, gene coverage score for the macrolide class demonstrated good predictive performance for erythromycin or clindamycin resistance with an AUC of 0.81 (95%CI: 0.62 – 0.99). Using the maximum Youden's index as the optimal cutoff, the sensitivity was 78% (95%CI: 61 – 95%) and specificity was 100%. Beta-lactam and fluoroquinolone gene coverage score did not adequately predict fluoroquinolone or oxacillin resistance pattern, respectively.

Discussion: Sequence-based AMR gene prediction highlights the diversity of resistance genes across major human pathogens in a prospective clinical cohort. Results demonstrate that these gene prediction algorithms hold potential for application in antibiotic susceptibility testing. Further evaluation is required to validate the performance of KARGA in a larger cohort with diverse resistance patterns.